



Charlie Embellisher

Software Engineer & Builder

✉ charlie.embellisher@example.com

[in/CharlieEmbellisher](https://github.com/CharlieEmbellisher)

[github](#)

Final-year MEng Computing (Machine Learning) student at Imperial College London with demonstrated expertise in full-stack development across Python, TypeScript, and React. Proven ability to deliver production-grade features in fast-paced startup environments, combining strong technical fundamentals with pragmatic problem-solving and stakeholder collaboration.

Experience

Software Engineering Intern — 2024 – Present

- Architected and deployed full-stack platform features using Python Flask, SQLAlchemy, and fastAPI to production environments.
- Designed optimised database schemas for instruction versioning and draft support, significantly improving read performance.
- Collaborated with sales team to develop rapid POCs for enterprise clients (HMRC, CGI), implementing bespoke API endpoints based on feedback.

Software Engineer at Company 2 — Oct 2018 – Present

- Engineered custom connectors for no-code platforms including Zapier, expanding product reach to non-technical users.
- Developed full-stack features utilising modern tech stack (React, Node.js, TypeScript) deployed to production systems.
- Optimised system architecture through schema design and performance tuning, delivering measurable improvements in application responsiveness.

Software Engineer at Company 3 — Aug 2019 – Present

- Built scalable backend services using Python and Node.js, supporting multi-agent retrieval-augmented generation workflows.
- Implemented version control and draft management systems, ensuring data integrity and enabling iterative development cycles.
- Led technical engagement with key stakeholders to translate business requirements into production-ready features and integrations.

Projects

Mobile Payment SDK — Oct 2020 – Oct 2024

- Engineered full-stack payment integration SDK using React and Node.js, enabling seamless transaction processing across platforms.
- Implemented secure authentication and encrypted data handling protocols, meeting PCI compliance requirements for production deployment.

E-commerce Microservices — Aug 2021 – Nov 2024

- Architected microservices infrastructure using Python and AWS, scaling to handle high-concurrency transactions efficiently.

- Designed and implemented CI/CD pipeline with Docker containerisation, reducing deployment time by 60% and improving reliability.

GraphQL API Gateway — Feb 2021 – Aug 2024

- Developed GraphQL API gateway using Node.js and TypeScript, consolidating multiple backend services into unified interface.
- Optimised query resolution and implemented intelligent caching layer, reducing latency by 45% and database load significantly.

Automated CI/CD Deployment Tool — Jun 2022 – Apr 2023

- Built end-to-end CI/CD automation framework using Docker and AWS, enabling one-command deployment across multiple environments.
- Implemented comprehensive logging and health-check mechanisms, reducing deployment failures by 80% and improving system reliability.

Machine Learning Data Pipeline — Oct 2021 – Apr 2024

- Developed scalable data pipeline using Python and AWS, processing and augmenting datasets for generative AI model training.
- Engineered automated data validation and transformation stages, ensuring 99.2% data quality for production ML workflows.

Education

BSc Computer Science — Sep 2014 – Oct 2020

Bristol, UK

Skills

React

Node.js

AWS

TypeScript

Python

JavaScript

Django